Muntasir Raihan Rahman

Principal Machine Learning Engineer | ML Infrastructure & Applied ML Research
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Summary

Principal ML engineer and researcher with 10+ years building production ML systems at Microsoft, Keystone AI, and Nokia Bell Labs. Currently hosting embedding and reranking models for Microsoft's Cosmos DB AI service, with deep expertise in inference optimization, distributed ML training, and large-scale model serving. PhD in distributed systems (UIUC); research record spanning ML systems, federated learning, and formal methods—most recently machine-checked guarantees (Lean 4) for agentic shared memory. Part-time faculty at UW Bothell, where I teach graduate machine learning.

Experience

Principal Engineer Apr 2026 – Present
Microsoft — Cosmos DB AI Service, Redmond, WA

- Cut reranker model GPU inference latency up to **15× on T4 and 4× on A100** using CUDA graphs; reduced tail latency via safe multi-processing for model inference.
- Design and ship embedding and reranking API features for Cosmos DB's AI service, hosting models used for search and memory workloads [SCALE: models/QPS served].
- Designed and prototyped scalable, work-conserving queue-based and asynchronous batch inference for reranking and embedding models.
- Strengthened inference reliability: improved telemetry, isolated worker pools, incident routing, and zero-downtime rollouts.
- Prototyped a multi-layer decision engine for memory retrieval and ingestion—cheap decision models at layer 1, expensive LLM calls at layer 2.
- Applied formal methods (Lean 4 proofs, Z3 SMT, model checking) to prove safety, security, and privacy guarantees for agentic shared memory systems.

Principal Machine Learning Engineer Feb 2024 – Mar 2026
Keystone AI — Core AI Group, Bellevue, WA

- Built foundation models for manufacturing supply chains on Keystone's Deep Enterprise AI Platform, a scientific decision engine integrated with enterprise ERP and planning systems to drive economically optimal decisions.
- Developed deep learning and foundation time-series forecasting models (DeepAR, Temporal Fusion Transformer, unified multi-client forecasting interface with ensembling and evaluation).
- Reinforcement learning and dynamic optimization models for scalable inventory placement and supply-chain optimization.
- ML engineering platform lead for a major Fortune 500 client: data processing, model building, and deployment across AWS and Azure, using Databricks and Snowflake.
- Research at the intersection of ML and econometrics under Dr. Patrick Bajari: auction econometrics, discrete choice modeling, and AI tooling for econometric libraries.

Principal Machine Learning Engineer Feb 2020 – Feb 2024
Microsoft — Copilot Applied AI / Dynamics 365, Redmond, WA

- Productionized a new LLM feature (NL2Journey) generating customer journeys from natural language for Power Platform / Dataverse enterprise customers.
- Built distributed DNN training infrastructure cutting model training time from ∼20 hours to ∼3 hours (10×); owned the applied ML team's end-to-end MLOps infrastructure.

- Deployed multiple production ML models at large scale, including deep neural network embedding models and unsupervised models for e-commerce and account-takeover fraud detection.
- Led data processing infrastructure for A/B testing and experimentation for large language models; built an ML performance evaluation framework with adaptive control of model hardware and parameters.
- New techniques for sentiment analysis using graph neural networks; scalable graph analytics and ML for fraud protection.
- Co-authored the device fingerprinting service paper (Siamese DNN over Bing's deep learning vector search service), *Microsoft Journal of Applied Research*, 2020.

Research Scientist Jan 2018 – Jan 2020
Nokia Bell Labs, Murray Hill, NJ

- Led research on ML for end-to-end service management and automation for Nokia's orchestration and assurance products; successful technology transfer to the business group.
- Fully decentralized, privacy-preserving federated learning: designed secure aggregation protocols without central coordination (published at FOGML/INFOCOM workshop, 2021).
- Deep reinforcement learning and mechanism design for resource allocation in 5G network slicing; lead cloud architect for the E2E 5G slicing testbed.
- Runtime verification and monitoring of end-to-end network services spanning SDN control plane and P4 data plane.

Software Engineer II Aug 2016 – Dec 2017
Microsoft — Azure, Redmond, WA

- Improved debugging capabilities of Cosmos, Microsoft's internal big-data analytics platform; designed a scale-out distributed log ingestion system using a distributed actor framework.
- Deployed log analytics systems to production data centers with AutoPilot; built end-to-end performance benchmarking tools.

Selected Publications

- Geo-distribution of Actor-Based Services. *Proc. ACM Program. Lang. (OOPSLA)*, 2017. (with Bernstein, Burckhardt, et al.)
- Characterizing and Adapting the Consistency-Latency Tradeoff in Distributed Key-Value Stores. *ACM Trans. Autonomous and Adaptive Systems*, 2017.
- SocialTrove: A Self-Summarizing Storage Service for Social Sensing. *IEEE ICAC*, 2015. **Best Paper Award**.
- OPTiC: Opportunistic Graph Processing in Multi-tenant Clusters. *IEEE IC2E*, 2018. **Best Paper Award**.
- Eventually Consistent: Not What You Were Expecting? *Communications of the ACM*, March 2014. (with Golab, AuYoung, Keeton, Li)
- Client-Centric Benchmarking of Eventual Consistency for Cloud Storage Systems. *IEEE ICDCS*, 2014.
- Formal Modeling and Analysis of Cassandra in Maude. *ICFEM*, 2014.
- Natjam: Design and Evaluation of Eviction Policies for Supporting Priorities and Deadlines in MapReduce Clusters. *ACM SoCC*, 2013.
- ViNEYard: Virtual Network Embedding Algorithms with Coordinated Node and Link Mapping. *IEEE/ACM Trans. Networking*, 2012.
- Privacy-Preserving Decentralized Aggregation for Federated Learning. *FOGML (INFOCOM workshop)*, 2021.

- A Scalable, Low-Latency Device Fingerprinting Service Powered by a Siamese DNN. *Microsoft Journal of Applied Research*, 2020.

Full list (30+ papers) on Google Scholar. Current research: certified program models for shared agentic memory—formal safety/security/privacy guarantees mechanized in Lean 4 and Rocq, with a verified Rust kernel evaluated on Azure Cosmos DB.

Education

- **Ph.D., Computer Science**, University of Illinois at Urbana-Champaign, 2016. Advisor: Indranil Gupta. Thesis: *Exploiting Cost-Performance Tradeoffs for Modern Cloud Systems*.
- **M.Math., Computer Science**, University of Waterloo, 2010.
- **B.Sc., Computer Science and Engineering**, Bangladesh University of Engineering and Technology, 2006. Ranked 5th in a class of 118.

Teaching

Part-time Faculty, University of Washington Bothell (2022 – Present): CSS 581 Machine Learning (graduate, Fall 2026), CSS 485 Artificial Neural Networks, CSS 448 Compilers, CSS 430 Operating Systems, CSS 421 Computer Hardware and Operating Systems, CSS 343 Data Structures and Algorithms, CSS 143 Computer Programming II.

Honors & Awards

Best Paper Awards, IEEE ICAC 2015 and IEEE IC2E 2018 | VMware Graduate Fellowship 2014–15 (one of 4 winners worldwide) | Feng Chen Memorial Award, UIUC, 2016 | CS Excellence Fellowship, UIUC | Dynamics 365 Products Group WMBB Award, 2021 | Microsoft Global Hackathon, 2022 and 2023.

Technical Skills

ML/AI: LLM serving and inference optimization (CUDA graphs, multi-process inference), embedding and reranking models, vector databases and vector search (Azure Cognitive Search), distributed DNN training (Horovod, Petastorm), foundation time-series models, reinforcement learning, graph neural networks, federated learning
Systems: distributed systems design, Azure and AWS, Databricks, Snowflake, Apache Spark, actor frameworks
Formal methods: Lean 4, Z3 SMT, TLA+, Maude, model checking
Languages: Python, C/C++, Rust, Java | **Languages (spoken):** English, Bengali